

Yue Zhan

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SUMMARY

3rd year Biostatistics Ph.D. student specializing in the development and application of **joint models**, **causal inference**, and **dynamic prediction** for complex clinical data. Seeking to leverage advanced statistical methods to solve challenges in **clinical trial** design and analysis. 1 year experience as graduate teaching assistant in SAS and basic biostatistics. 1 year experience in **consulting** and **collaborating** across multi-disciplinary teams to analyze their data and solve statistical problems.

EDUCATION

University of Nebraska Medical Center

Omaha, NE

Ph.D. in Biostatistics, GPA: 3.97/4.00, Qualify Exam Passed

Aug 2023 – Present

Thesis: **Joint analysis** for biomarkers with a **change point** anchored at **interval-censored** event

Curriculums: Survival Analysis, Correlated Data Analysis, Bayesian Theory and Method, Causal Inference, Theory of Generalized Linear/Mixed Models, Biostatistics Consulting Application/Practice, Advanced Biostatistics Theory I/II

Shanghai Jiao Tong University

Shanghai, China

B.S. in Mathematics and Applied Mathematics, GPA: 3.68/4.30

Sept 2019 – Jun 2023

Thesis: **Temporal Difference Algorithm's Implicit Bias and Continuous Modeling**

Curriculums: Probability Theory, Mathematical Statistics, Stochastics Process, Statistical Learning, Convex Optimization

RESEARCH EXPERIENCE

University of Nebraska Medical Center

Omaha, NE

Graduate Research Assistant

Aug 2023 – Present

Dissertation: **Joint analysis** for multivariate biomarkers with a **change point** anchored at **interval-censored** event time, with applications to Huntington's Disease (HD).

- **Topic 1:** Develop a **two-phase joint model** to handle the change point of biomarkers and interval-censored data. Test two hypotheses: cognitive decline predicts HD onset and HD onset accelerates cognitive decline.
 - Propose a **two-phase joint model** with biomarkers' trajectories change after an interval-censored event time.
 - Use two decompositions of likelihood to represent the likelihood of this two-phase joint model.
 - Use **B-spline** to approximate the cumulative hazard and use Newton-Raphson algorithm to solve the MLE.
 - Simulations by **R** show good performance of model and consistency of MLE.
 - Application on PREDICT-HD data to see the effects of cognitive biomarkers on HD onset and vice versa.
- **Topic 2:** Develop **causal mediation** analysis on two causal pathways based on the two-phase joint model. Conduct **dynamic prediction** of event time and trace of biomarkers on individual level given trace of biomarkers in part of the first phase.
 - Use g-formula to represent the expectation of biomarkers/survival functions with given covariate's value.
 - Compute NDE and NIE of given covariate on biomarkers/survival functions through its mediator. Simulations are conducted to demonstrate its performance.
 - Application on real data shows the causal effect of CAP (HD disease burden) on biomarkers/survival functions.
 - Derive joint distribution of biomarkers after a fixed time and event time given biomarkers' trace before this fixed time. Predict event time and biomarkers' trajectory by this distribution.

Shanghai Jiao Tong University

Shanghai, China

Undergraduate Dissertation: **Temporal Difference Algorithm's Implicit Bias and Continuous Modeling**

Sept 2022 – Jun 2023

- Explore the implicit regularization and convergence property of temporal difference algorithm in **reinforcement learning** through simulation using **Pytorch in Python**.
 - Analyze the convergence properties of temporal difference algorithm in a simple environment setting.
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WORK EXPERIENCE

University of Nebraska Medical Center

Omaha, NE

Graduate Research Assistant

Aug 2025 – Present

- ProjectDRIVE: A **randomized trial** to improve safe driving behaviors in teens with traffic violations.
 - Manage and clean data, identify **missing values** for a randomized trial to ensure data integrity for analysis.

- Define a novel communication score to quantify parent-teen interaction as a key variable.
- Use **generalized linear model** to study the relationship between parent-teen communication and risky driving/unsafe behaviors/recidivism.

Center for Collaboration on Research Design and Analysis (CCORDA)

Omaha, NE

Graduate Research Assistant

Aug 2024 – Jun 2025

- Provide **statistical consulting** to researchers from diverse fields such as Dermatology and Oncology about study design, sample size calculation and test selection, translating interdisciplinary research questions into statistical analysis plans.
- Analyze their data and interpret results to them, mainly including **data cleaning, descriptive analysis, comparison** among groups and **regression** analysis using **SAS and R**.
- Part of projects list:
 - i) Reported vs. Actual Patient Data (Dermatology)
 - Use **Wilcoxon signed rank test** to compare difference between the actual and patient-reported number of skin cancers and actinic keratoses at 6 and 12 months.
 - Use **McNemar's test** to compare difference between chemoprevention prescribed by the physician and patient-reported chemoprevention use at 6 and 12 months.
 - Use **T-test** to compare difference between the patient's actual and expected follow-up visit date with their general dermatologist post-Mohs surgery.
 - ii) Prognostic Factor in Older Adults with Acute Myeloid Leukemia (AML) Participating in a Geriatric Assessment (GA)
 - Use **log-rank test** to examine associations between mortality and impairments in GA.
 - Use **Cox model** after adjusting for covariates such as age, WBC count, and AML risk categories.
 - iii) Histogram Analysis for MD Ratios in MRI Data (Radiology)
 - Descriptive analysis for each lesion within all four MRI Types (HGG/Benign × Patient/Control).
 - Comparison tables for descriptive statistics and **Fisher's exact** test for categorized skewness and kurtosis between HGG and Benign groups in patient and control groups.
 - **ROC Curve** Analysis to determine which statistic is best for differentiating Benign from HGG tumors.
 - iv) Atopic Dermatitis ED Project (Dermatology)
 - Use **SQL** to combine datasets from different sites, unify variables' names, check missing data.
 - Use **Pearson correlation, ANOVA** and **T-test** to compare the difference of LOS (length of stay) in ED across different covariates' groups.

University of Nebraska Medical Center

Omaha, NE

Graduate Teaching Assistant

Aug 2023 – May 2024

- Assistant teaching in class, help students understand the contents of class and give feedback to their homework. Answer students' questions in office hours. Communicate with students and professor to let the class progress with appropriate speed.
- Teaching assistant in Intro to SAS in fall 2023 semester. Master based and advanced SAS programming skills. Teaching assistant in Biostatistics II In spring 2024 semester.

PUBLICATIONS

- Aby, E. S., Pradhan, F., King, L. Y., Kuo, A., Fox, A., Norvell, J. P., ..., **Zhan, Y.** & Pillai, A. Workload, compensation, and institutional support among US transplant hepatologists: Results of a National Survey. *Liver Transplantation*, 10-1097.
- Fagan, E., McGrath, J., Samson, K., **Zhan, Y.**, & Sutton, A. (2025). Discrepancy Between Patient-Reported and Physician-Prescribed Chemoprevention in Patients With Keratinocyte Carcinomas. *International Journal of Dermatology*.

TECHNICAL SKILLS

- **Programming Languages:** R (ggplot2, shiny, survival, dplyr), SAS (Macro, SQL), Python (TensorFlow, Pytorch)
- **Tools:** SQL, SPSS, R Markdown, Microsoft Office Suite, Parallel Computing
- **Certification:** SAS Base Programmer, SAS Advanced Programmer

PRESENTATIONS

- Invited speaker at Lifetime Data Science 2025 in Brooklyn, NY and International Indian Statistical Association 2025 in Lincoln, NE: **Joint analysis** for multivariate longitudinal and event time data with a change point anchored at interval-censored event time